

Following an investigation, the head of the data science function has suggested four reasons for the model underperformance. These are class imbalance, data quality issues, overfitting, and model drift.

4 Risk concerns

The CRO has recently seen press coverage concerning a competitor, FernLink Co.

The share price of FernLink Co has experienced a very sharp decline. The share price decline has been attributed to significant financial losses and reputational damage. The financial losses relate to an increase in the volume of fraudulent transactions incurred. FernLink Co has also suffered reputational damage following a regulatory investigation into its fraudulent transactions and a spike in customer complaints due to the number of legitimate transactions flagged as suspicious. It is suggested that governance of their fraud prediction models has contributed significantly to this.

The CRO wishes to understand how EP can mitigate against similar issues.

5 Appendix 1

	A	B	C
1	Column	Type	Description
4	customer_id	String	Synthetic customer identifier.
5	customer_age	Integer	Customer age in years
6	customer_tenure_days	Integer	Days since customer signup
7	is_vip	Binary (0/1)	1 if customer is VIP / high-value segment.
8	merchant_id	String	Synthetic merchant identifier.
9	merchant_category	Categorical	Merchant sector/category (e.g., grocery, electronics, travel, etc.).
10	merchant_risk_score	Float (0–1)	Merchant risk propensity (higher = riskier).
11	amount	Float	Transaction amount (positive; lognormal-like).
12	currency	Categorical	Transaction currency (EUR/GBP/USD).
13	payment_method	Categorical	Payment rail (card/bank_transfer/wallet).
14	channel	Categorical	Origin channel (web/mobile_app/pos; may include in_app in production).
15	card_present	Binary (0/1)	1 if card-present (mostly POS).
16	device_type	Categorical	Device OS (ios/android/windows/macOS/linux).
17	device_fingerprint	String / Null	Pseudo device identifier; may be missing in production.
18	billing_country	Categorical	Billing country code.
19	shipping_country	Categorical	Shipping country code (may differ from billing).
20	ip_country	Categorical	IP geolocation country code
21	is_cross_border	Binary (0/1)	1 if billing/shipping/IP countries are inconsistent (proxy for cross-border risk).
22	txn_count_1h	Integer	Number of transactions for customer in last 1 hour (velocity feature).
23	txn_count_24h	Integer	Number of transactions for customer in last 24 hours (velocity feature).
24	prev_chargebacks_12m	Integer	Number of prior chargebacks in last 12 months.
25	email_domain	Categorical	Customer email domain (e.g., gmail.com, business.com, student.edu, ...).
26	is_new_device	Binary (0/1)	1 if device appears new for this customer (proxy).
27	actual_fraud	Binary (0/1)	Target label: 1 if fraudulent, 0 otherwise.

6 Appendix 2

Refer to the spreadsheet titled SDS Data Science Professional Specimen Examination 1 – Question 1 Appendix 2